

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A — Code of Conduct

I __**PERAM VIVEKANANDA REDDY(192372330)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Vivek*

Annexure A — Article Review (Question Paper)

Instructions

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Article	20%	Clear and in-depth understanding of the article's purpose, concepts, and arguments	Good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Summary	20%	Concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Insightful critique with strong reasoning and evaluation	Reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ARTICLE REVIEW

Article Selected

N. L. Fitriyani, M. Syafrudin, G. Alfian, and J. Rhee, "HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System," IEEE Access, vol. 8, pp. 133034–133050, 2020.

DOI: <https://doi.org/10.1109/ACCESS.2020.3010511> (open access, IEEE Xplore, Creative Commons Attribution 4.0 License). As of 2024, this article had accumulated 252 Scopus-indexed citations, indicating strong peer recognition.

Task 1: Article Summary

(a) Citation, Domain & ML Problem

Full citation:

Fitriyani, N. L., Syafrudin, M., Alfian, G., & Rhee, J. (2020). HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System. IEEE Access, 8, 133034–133050. DOI: 10.1109/ACCESS.2020.3010511.

Domain / application area:

Healthcare informatics — specifically, clinical decision support systems (CDSS) for early diagnosis of cardiovascular disease (heart disease).

Specific ML problem addressed:

Binary classification: predicting the presence or absence of heart disease in a patient from clinical/demographic attributes (age, sex, chest pain type, blood pressure, cholesterol, ECG results, maximum heart rate, etc.), while explicitly tackling two data-quality problems that commonly degrade classifier performance on medical tabular data — outlier contamination and class imbalance between diseased and healthy patients.

(b) Key Objective, Research Gap & Approach (150–200 words)

The authors aim to build a more reliable heart-disease prediction model for real clinical use by directly confronting two data-quality problems that most prior studies overlooked: noisy outlier records and imbalanced class distributions between diseased and healthy patients. Their central claim is that earlier heart-disease prediction studies had compared various classifiers (Naïve Bayes, SVM, neural networks, ensembles, and so on) but had not combined density-based outlier removal with a hybrid over/under-sampling balancing step ahead of a strong gradient-boosted tree classifier. To close this gap, the authors designed a three-stage pipeline — DBSCAN-based outlier detection and removal, SMOTE-ENN hybrid class balancing, and an XGBoost classifier — and validated it on two well-known public datasets (Statlog and Cleveland), benchmarking it against six standard classifiers and against numbers reported in prior published studies. They further embedded the resulting model into a working web-based clinical decision support system prototype, showing the work was intended for practical deployment, not purely academic benchmarking.

Task 2: Methodology Analysis

(a) ML Technique(s) Used & Dataset Description

ML techniques:

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise): an unsupervised density-based clustering algorithm used here purely to flag and remove outlier patient records before training, based on a 5-nearest-neighbour distance heuristic to choose the neighbourhood radius (eps).
- SMOTE-ENN: a hybrid class-balancing technique combining SMOTE (Synthetic Minority Over-sampling Technique), which generates new synthetic minority-class samples by interpolating between a real minority sample and its nearest minority neighbours, with Edited Nearest Neighbour (ENN) under-sampling, which then removes ambiguous/overlapping samples near the class boundary.
- XGBoost (Extreme Gradient Boosting): the core classifier — an ensemble of boosted decision trees that sequentially fits new trees to the residual errors of prior trees, regularised to control tree complexity and prevent overfitting. This is fundamentally a decision-tree-based inductive learning method, directly extending the single decision tree taught in CO4 into a much stronger ensemble form.
- Baselines for comparison: Naïve Bayes, Logistic Regression, Multilayer Perceptron, SVM, a single Decision Tree, and Random Forest — giving a broad comparison across statistical, neural, and tree-based inductive learners.

Dataset description:

Property	Dataset I — Statlog	Dataset II — Cleveland
Source	UCI Machine Learning Repository	V.A. Medical Center, Long Beach & Cleveland Clinic Foundation (via UCI)
Original size	270 subjects	303 subjects (76 raw attributes)
Used size (after cleaning)	270 subjects (no missing values)	297 subjects (6 removed for missing values)
Features used	13 clinical/demographic attributes	13 of the 76 available attributes
Class distribution (original)	120 disease-positive / 150 disease-negative	137 disease-positive / 160 disease-negative (after binarising a 0–4 severity scale)

Data preparation involved: (i) Pearson correlation and Information-Gain-based feature screening in Weka, which dropped weakly-correlated attributes (cholesterol, resting blood pressure, fasting blood sugar) from both datasets; (ii) DBSCAN outlier removal (with MinPts=5, eps=9 for Statlog and eps=8 for Cleveland, chosen via a 5-NN 'knee' heuristic); (iii) SMOTE-ENN rebalancing of the training folds, raising the minority-class share from 44.19%→50.79% (Statlog) and 46.05%→49.5% (Cleveland); and (iv) 10-fold cross-validation for every model to obtain averaged performance metrics and to reduce overfitting risk.

(b) Mapping to CO4 Topics, Strength & Limitation

Mapping to CO4 topics:

- Decision Trees: XGBoost is an ensemble/boosting extension of the decision tree algorithms studied in CO4 (Gini/Information-Gain-based splits at the level of each individual weak-learner tree).
- Inductive Learning: The overall approach — learning a general classification rule from labelled historical patient records to predict outcomes for new patients — is a textbook supervised inductive learning setup.
- Statistical Learning: Naïve Bayes and Logistic Regression are included as baseline statistical learners for comparison, directly relevant to the CO4 statistical-learning-methods topic, even though they are not the paper's proposed technique.

- Reinforcement Learning: Not used in this article — the paper's contribution sits entirely within the Decision-Trees/Inductive-Learning/Statistical-Learning topics of CO4, with no sequential decision-making or reward-based component.

Methodological strength:

The authors do not simply propose a new classifier in isolation — they run a systematic ablation showing the incremental contribution of each pipeline stage (they report that DBSCAN-based outlier removal alone improved accuracy from roughly 80.7%/80.0% to 85.4%/85.3% on the two datasets before SMOTE-ENN or XGBoost were even applied), and they confirm the final performance gap over baseline models using a paired t-test (rejecting the null hypothesis of no difference at $p < 0.05$ for every comparison). This turns an otherwise ordinary 'our model beats their model' comparison into a more rigorous, componentwise-justified result.

Limitation:

The authors themselves acknowledge that both datasets represent a specific demographic (largely U.S.-based clinical populations), that the model has not been evaluated in an actual clinical trial, and that no feedback from practising cardiologists had yet been obtained at the time of publication. Combined with the very small absolute sample sizes (270 and 297 patients), this significantly limits how far the reported performance can be expected to generalise to other populations, healthcare systems, or real deployment settings.

Task 3: Results & Critical Evaluation

(a) Key Results

The authors report eight performance metrics per model (accuracy, precision, recall/sensitivity, F-measure, MCC, AUC, false-positive rate, true-negative rate) via 10-fold cross-validation. The headline results for the proposed HDPM (DBSCAN + SMOTE-ENN + XGBoost) were:

Metric	Dataset I — Statlog	Dataset II — Cleveland
Accuracy	95.90%	98.40%
Precision	97.14%	98.57%
Recall / Sensitivity	94.67%	98.33%
F-measure (F1)	95.35%	98.32%
Matthews Correlation Coefficient (MCC)	0.92	0.97
AUC-ROC	1.00	1.00
False Positive Rate	4.52%	1.67%
True Negative Rate (Specificity)	95.48%	98.33%

The proposed HDPM outperformed all six baseline classifiers (NB, LR, MLP, SVM, DT, RF) on every one of the eight metrics for both datasets, and the difference was confirmed statistically significant via paired t-tests ($h=1$, $p<0.05$, calculated $t >$ tabulated t for every comparison). The authors also benchmarked against seven previously published studies using the same two public datasets, reporting an average accuracy improvement of 8.12% (Statlog) and 9.83% (Cleveland) over those prior results.

(b) Critical Evaluation of the Results

Are the reported metrics realistic?

- The accuracy and F1 figures (95.9%–98.4%) are high but broadly consistent with other published results on these same two small, well-studied benchmark datasets, so they are not implausible on their face.
- However, an AUC-ROC of exactly 1.00 on both datasets is a genuine red flag. A perfect AUC means the model achieved perfect rank-ordering of every diseased vs. healthy patient in cross-validation — an extremely strong claim for real clinical data, and one that invites scrutiny of whether any information leakage occurred (for example, whether SMOTE-ENN's synthetic oversampling was generated using neighbour information that inadvertently spanned across a fold boundary, or whether outlier removal was performed before rather than within each cross-validation fold). The paper's methodology section does not give enough procedural detail to fully rule this out, which is a common but under-scrutinised risk in papers using SMOTE within cross-validation.

Potential biases:

- Dataset bias: Both datasets are small (270 and 297 patients) and drawn from specific clinical sites in a single country, which the authors themselves flag as limiting generalisability to other demographics.
- Evaluation-protocol bias: With sample sizes this small, 10-fold cross-validation folds contain only around 27–30 patients each, so metric estimates (especially AUC and MCC) can have high variance, and a handful of easy or hard cases can disproportionately swing the reported score.
- Synthetic-data bias: SMOTE-ENN generates synthetic minority patients by interpolation; if these synthetic points are too close to real patients that later appear in a validation fold, the model can be evaluated on data that is not truly independent of its training set, artificially inflating performance.

Additional experiments that would strengthen the findings:

- External validation on a held-out dataset from a different country/hospital system, entirely unseen during model development, to test genuine generalisability.
- An explicit leakage-safe cross-validation protocol (applying DBSCAN, feature selection, and SMOTE-ENN inside each training fold only, never touching the held-out fold) with the code released for reproducibility.
- Reporting confidence intervals or standard deviations across the 10 folds for every metric, rather than single point estimates, given the small sample sizes.
- A prospective clinical validation study with practising cardiologists — which the authors explicitly list as future work — to test real diagnostic utility rather than retrospective benchmark accuracy.

(c) Real-World Applicability & Scaling Challenges

The authors already moved beyond pure benchmarking by designing and implementing a working prototype — the Heart Disease Clinical Decision Support System (HDCDSS) — a web application where clinicians enter a patient's current clinical data and receive an instant presence/absence diagnosis, with data stored in MongoDB on a local (non-cloud) server to preserve patient confidentiality. This demonstrates a credible deployment pathway: hospital-based or primary-care CDSS tools that give clinicians a fast, evidence-based second opinion alongside traditional diagnostic methods (ECG, angiography).

Challenges in scaling to industry:

- **Data availability:** Real hospitals would need to collect substantially larger and more demographically diverse patient cohorts than the 270–297-patient benchmark datasets used here before the model could be trusted across different populations.
- **Compute:** The pipeline itself (DBSCAN + SMOTE-ENN + XGBoost, all trained via CPU-only libraries) is computationally lightweight — the authors note it ran on an ordinary 8-core desktop — so compute is not a major scaling barrier compared to deep-learning-heavy alternatives.
- **Ethical issues:** A false negative in this context means a genuinely diseased patient is told they are healthy, with potentially fatal consequences if it delays treatment; deployment would require careful threshold tuning toward higher sensitivity, human-in-the-loop clinical oversight (which the authors' own CDSS design supports), regulatory approval as a medical device, and ongoing bias auditing across age, sex, and ethnicity before wider rollout — none of which the current study addresses.
- **Local-network privacy design:** The authors deliberately kept patient data off the cloud and on a local hospital server, which is a sound design choice for compliance with medical data-protection regulations, but limits the system's ability to pool data across multiple hospitals to grow its training set, creating tension between privacy and generalisability.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Connected to CO4

Insight 1 — Ensemble decision trees extend, rather than replace, single-tree reasoning:

Seeing XGBoost dramatically outperform a single Decision Tree baseline in the same paper made concrete something CO4 introduces conceptually: that boosting many shallow trees sequentially on residual errors trades some interpretability for a large reduction in variance and bias compared with one deep tree, directly extending the CO4 Decision-Tree-algorithms topic into ensemble learning.

Insight 2 — Data preprocessing can matter as much as the choice of algorithm:

The paper's own ablation, showing accuracy jump from ~80% to ~85% purely from DBSCAN outlier removal before any classifier change, reinforced the CO4 inductive-learning principle that a learner's inductive bias is only as good as the data it inducts from — noisy or imbalanced training data can dominate algorithm choice as a source of error.

Insight 3 — Statistical learners remain a meaningful comparison baseline, not a strawman:

Although Naïve Bayes and Logistic Regression were outperformed here, their inclusion as principled statistical-learning baselines (rather than being omitted) reflects the CO4 statistical-learning-methods topic's emphasis on comparing model families honestly — a reminder that the 'best' model is context- and dataset-dependent, not universally the most complex one.

(b) Proposed Extension Experiment

Proposed experiment:

Re-run the authors' full DBSCAN → SMOTE-ENN → XGBoost pipeline with a strict, leakage-safe nested cross-validation protocol (all preprocessing fit only within each training fold) on a larger, multi-country heart-disease cohort — for example, combining the Statlog and Cleveland datasets with the additional Hungarian, Switzerland, and Long Beach VA heart-disease datasets already hosted in the same UCI repository family — and add a SHAP (SHapley Additive exPlanations) explainability layer on top of the final XGBoost model.

Justification of feasibility:

- All four additional UCI heart-disease datasets share a highly similar 13/14-attribute schema to Statlog/Cleveland, so integrating them requires only modest data-harmonisation effort, not new data collection.
- SHAP is a mature, widely used, open-source library that works natively with XGBoost with minimal extra implementation effort, and would directly address the paper's own acknowledged interpretability/clinical-trust gap without needing to change the underlying model.

Expected impact:

- Pooling multiple demographically distinct cohorts would give a much more credible test of generalisability than the original single-country evaluation, directly answering the authors' own stated limitation.
- A leakage-safe nested cross-validation would produce a more trustworthy (likely somewhat lower, but more honest) AUC estimate, addressing the critical-evaluation concern raised in Task 3(b) about the suspiciously perfect AUC of 1.00.
- Adding SHAP-based per-patient explanations would give clinicians a transparent rationale for each prediction (e.g., 'flagged high-risk primarily due to elevated ST-depression and low maximum heart rate'), which is essential for real clinical adoption and directly strengthens the ethical/explainability gap identified in Task 3(c), at a relatively low additional engineering cost.